

K. Vaikunth Keshav

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Education

M.S. in Mechanical Engineering, University of Colorado Boulder **Expected Graduation:** May 2026
Cumulative GPA: 3.71/4.0

Relevant Courses: Design for Manufacturability, Advanced Robotics, Advanced Computer Vision, Advanced Product Design, Computational Fluid Dynamics, Mechatronics-I, Introduction to Robotics, Methods of Engineering Analysis 1

B.E. in Mechanical Engineering, Minor in Finance, BITS Pilani-Dubai Campus, UAE **Graduated:** 2021
GPA: 7.36/10.0

Technical Skills

Mechanical Engineering: Mechanical Design, P&ID development, CAD, DFM, DFA, GD&T, Quality Assurance

3D CAD Design Software: SolidWorks, Solid Edge, CATIA, AutoCAD, Revit, Creo, NX CAD, Navisworks

FEA Analysis Tools: Ansys, Abaqus

Programming: ROS 2, Python, C++

Computer Vision & Robotics: ROS2, Arduino, SLAM, OpenCV, PyTorch Deep Learning, U-Net

Productivity Software: Microsoft Excel, Microsoft Word, Microsoft Office, BlueBeam Revu

Professional Experience

Engineer – Mechanical Design **June 2022 – Sep 2023**
Hamsa Environmental Solutions Private Limited, Tamil Nadu, India

- Drafted 2D designs for heat exchangers, pressure vessels, and WFEs in AutoCAD, ensuring ANSI B16 compliance for chemical waste treatment systems including used oil re-refinery and solvent extraction plants.
- Developed 3D WFE assemblies in Solid Edge, improving design accuracy and reducing errors in manufacturing.
- Designed structural skid layouts and foundation drawings using standardized load calculations for processing equipment.
- Progressed to process engineering work, developing P&ID drawings in AutoCAD for UORS, LPS, and SES installations.
- Utilized PV-Elite for pressure vessel FEA to validate structural integrity and prevent manufacturing failures, and CADWorx for piping and plant layouts.

Intern – Mechanical Design **Jan 2021 – July 2021**
Subzero Motorsport, Dubai, UAE

- Assisted with vehicle ECU integration, diagnostics, and supported CAE software usage.

Intern – Mechanical Design **June 2019 – Aug 2019**
Al Al Ashwaq Building Maintenance, Electrical Contracting and Plumbing LLC, Dubai, UAE

- Designed 2D layouts for electrical, plumbing, HVAC, and firefighting systems in AutoCAD.

Projects

Autonomous Hallway Navigation, University of Colorado Boulder **Aug 2025 - Dec 2025**
Built autonomous ground robot using LIDAR-based SLAM with offline mapping and online waypoint navigation. Completed visual line following and reverse driving challenges. Implemented all navigation logic, control algorithms, and waypoint management from scratch.

Optimizing Heat Transfer in Heat Sink using CFD, University of Colorado Boulder **Jan 2025 - May 2025**
Investigated optimal fin configuration for passive heat sinks using computational fluid dynamics. Simulated natural convection in three fin array configurations, determining that 10-fin design achieved 62% higher heat transfer than 6-fin and 18% higher than 13-fin configurations.

Reverse Engineering of a Handheld Wireless Vacuum, University of Colorado Boulder **Aug 2024 - Dec 2024**
Reverse-engineered and redesigned vacuum for manufacturability, reducing part count by 46%. Conducted material selection using Ashby charts, DFA analysis, and component evaluation for improved assembly and cost reduction.

Chocolate Winnowing Machine Design, University of Colorado Boulder **Aug 2024 - Dec 2024**
Developed a novel cacao winnowing machine with integrated vacuum, reducing part count from 73 to 37 through DFA analysis and achieving a unit cost of \$441.80.

Smart Concussion Detection Helmet, University of Colorado Boulder **Jan 2025 - Apr 2025**
Developed prototype smart helmet using Arduino UNO R4, accelerometer sensors, and WiFi connectivity to detect and transmit real-time head impact data. Designed mobile app interface via MIT App Inventor and established cost-effective manufacturing strategy targeting the 9.2M U.S. winter sports market.

Mario Kart Balloon Battle, University of Colorado Boulder **Aug 2024 - Dec 2024**
Collaborated to engineer an autonomous robot with mecanum wheels, sensors, and a PixyCam for object detection in a Mario Kart Balloon Battle.

4-DOF Robotic Arm with Gripper, University of Colorado Boulder **Aug 2024 - Dec 2024**
Simulated a robotic arm with a gripper, utilizing forward kinematics for motion control in Solidworks.

Autonomous Hallway Navigation, University of Colorado Boulder

Aug 2024 - Dec 2024

Implemented autonomous ground robot navigation system using LIDAR-based SLAM with offline mapping and online waypoint following. Built two-mode SLAM system: mapping mode using SLAM Toolbox for occupancy grid generation, and planning mode with localization-only operation. Developed PD waypoint controller with dual-criteria advancement logic. Completed visual line following challenge using RGB camera with proportional control, and reverse driving challenge using LIDAR. Implemented all navigation logic, control algorithms, and waypoint management from scratch. Achieved average lap time of 1:10 with 6.24s standard deviation over 13 trials.

Monocular Visual Odometry, University of Colorado Boulder

Oct 2024 - Nov 2024

Implemented complete monocular visual odometry pipeline on KITTI Sequence 00. Developed optical flow-based feature tracking using Lucas-Kanade algorithm, essential matrix estimation with RANSAC outlier rejection, and incremental 6-DOF trajectory integration over 1000 frames. Applied sliding window bundle adjustment for local pose refinement. Achieved Absolute Trajectory Error (ATE) of 9.95m (~1% drift) and maintained 200-400 features per frame with 5-15 frame lifetimes.

Structure-from-Motion & Camera Pose Estimation, University of Colorado Boulder

Sep 2024 - Oct 2024

Developed incremental SfM pipeline for camera pose estimation and sparse 3D reconstruction from unordered images. Implemented 8-point algorithm with RANSAC for fundamental matrix estimation, essential matrix decomposition for pose recovery, linear and nonlinear triangulation for 3D point reconstruction, PnP RANSAC for camera pose estimation, and bundle adjustment for global refinement. Applied to Lego (100 images), Dino (90), and Buff (158) datasets achieving accurate 3D reconstructions.

Coded Deep Depth Estimation, University of Colorado Boulder

Oct 2024 - Nov 2024

Implemented deep-learning pipeline for monocular depth estimation using coded images generated via depth-dependent PSF convolution. Designed and trained U-Net architecture with encoder-decoder structure on NYUv2 and UMD Coded VO datasets. Achieved 70% reduction in absolute relative error compared to AiF models (0.0610 vs 0.2108 for Dining Room) while maintaining real-time inference speeds of ~880 FPS. Implemented depth-weighted loss function for improved training.

Panoramic Image Stitching Pipeline, University of Colorado Boulder

Sep 2024

Developed complete Python-based computer vision pipeline using OpenCV for panoramic image stitching across 8 diverse datasets. Integrated Harris corner detection, ANMS for feature selection, 64-dimensional feature descriptors, SSD-based feature matching, RANSAC-based homography estimation, and cylindrical warping to generate seamless multi-image panoramas. Implemented custom warp and blend algorithms for smooth transitions between images.

Camera Calibration using Zhang's Method, University of Colorado Boulder

Aug 2024 - Sep 2024

Implemented Zhang's camera calibration method to estimate intrinsic parameters (focal lengths, principal point) and radial distortion coefficients from checkerboard images. Computed homographies, solved for intrinsic matrix using closed-form solution, applied nonlinear optimization for refinement, and performed lens distortion correction with reprojection error minimization.

Certifications

- **Product Design (SolidWorks, AutoCAD at CADD Centre).**

June 2018 - Dec 2020